2. Write the formula for the following coordination compounds.

   a) potassiumhexacyanidoferrate(II)

   b) pentacarbonyliron(0)

   c) pentaamminenitrito $-\kappa-N$ -cobalt(III)ion

   d) hexaamminecobalt(III)sulphate

   e) sodiumtetrafluoridodihydroxidochromate(III)

3. Arrange the following in order of increasing molar conductivity

   i) $Mg[Cr(NH_3)(Cl)_5]$ ii) $[Cr(NH_3)_5Cl]_3[CoF_6]_2$

   iii) $[Cr(NH_3)_3Cl_3]$

4. Give an example of coordination compound used in medicine and two examples of biologically important coordination compounds.

5. Based on VB theory explain why $[Cr(NH_3)_6]^{3+}$ is paramagnetic, while $[Ni(CN)_4]^{2-}$ is diamagnetic.

6. Draw all possible geometrical isomers of the complex $[Co(en)_2Cl_2]^+$ and identify the optically active isomer.

7. $[Ti(H_2O)_6]^{3+}$ is coloured, while $[Sc(H_2O)_6]^{3+}$ is colourless- explain.

8. Give an example for complex of the type $[Ma_2b_2c_2]$ where a, b, c are monodentate ligands and give the possible isomers.

9. Give one test to differentiate $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$.

10. In an octahedral crystal field, draw the figure to show splitting of d orbitals.

11. What is linkage isomerism? Explain with an example.

12. Classify the following ligand based on the number of donor atoms.

    a) NH_3 b) en c) ox^{2-} d) pyridine

13. Give the difference between double salts and coordination compounds.

14. Write the postulates of Werner's theory.

15. Why tetrahedral complexes do not exhibit geometrical isomerism.

16. Explain optical isomerism in coordination compounds with an example.

17. What are hydrate isomers? Explain with an example.

18. What is crystal field splitting energy?

19. What is crystal field stabilization energy (CFSE) ?

20. A solution of $[Ni(H_2O)_6]^{2+}$ is green, whereas a solution of $[Ni(CN)_4]^{2-}$ is colorless - Explain

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